FACULTÉ
DE MÉDECINE
UM6SS - CASABLANCA

SEMIOLOGY OF DIABETES MELLITUS

Pr IOANA HRISTOV

Semiology- Endocrinology 1

2026-2027

UNIVERSITÉ MOHAMMED VI
DES SCIENCES DE LA SANTÉ
CASABLANCA

www.um6ss.ma

I- DIABETES- LEARNING OBJECTIVES

- List the signs that should lead to the suspicion of type 1 or type 2 diabetes and distinguish between these two types of diabetes
- Explain the acute complications of diabetes
- Explain the chronic complications of diabetes

Diabetes definition

INTRODUCTION

Definition = chronic hyperglycemia due to an absolute or relative deficiency of insulin secretion and/or insulin action (insulin resistance (IR)).

- Public health issue
- Extremely common condition
- 2 main types: **type 1 diabetes**: autoimmune origininsulinopenia

  type 2 diabetes: insulin resistance
- Serious disease, risk of acute and chronic complications
- High morbidity and mortality

normal regulation of blood glucose:

- Fasting blood glucose < 1,10 g/l

- Blood Glucose < 1,40g/l 2 hours after OGTT

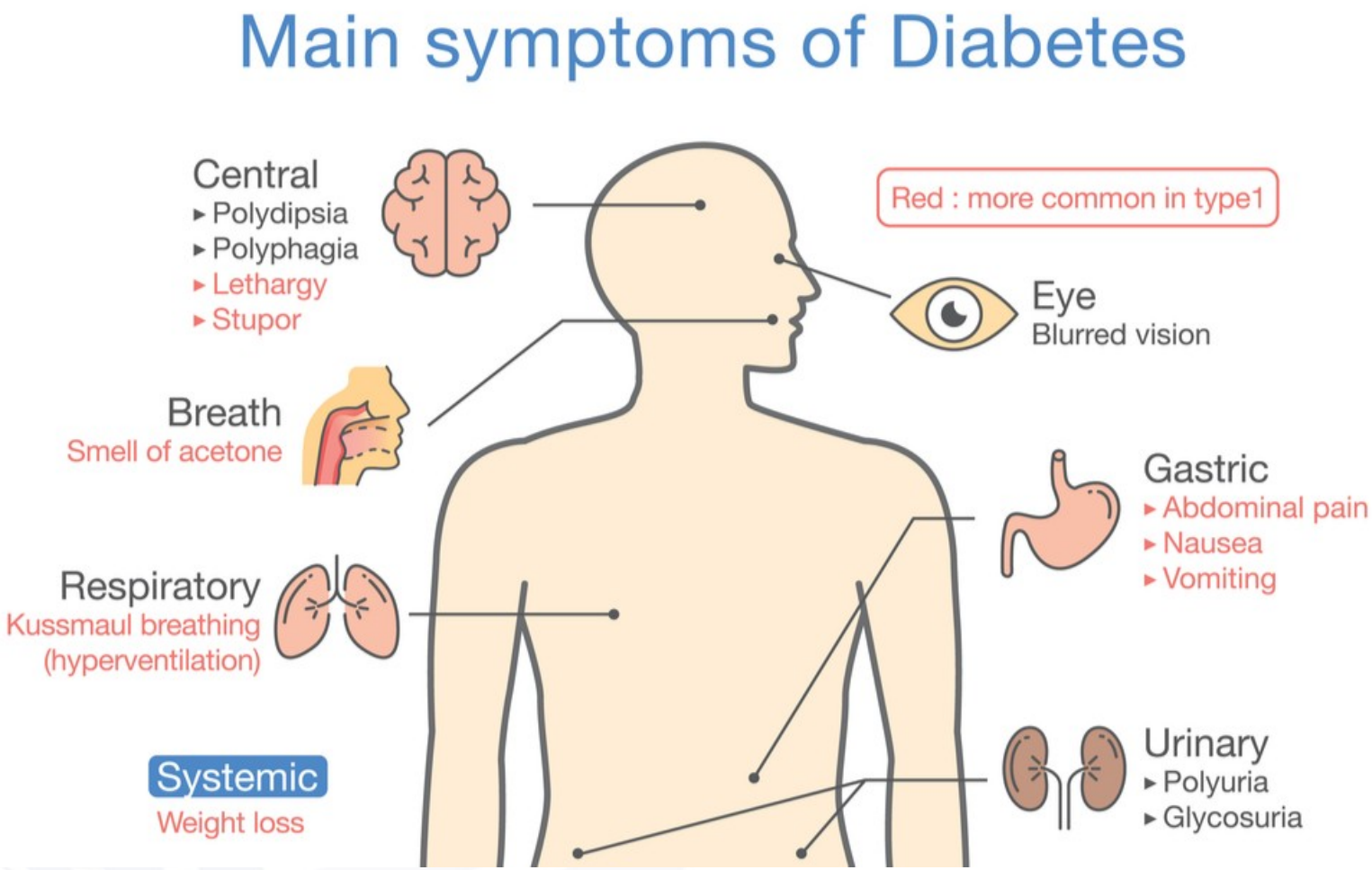
Clinical Presentation of Diabetes Mellitus

Diagnosis circumstance :

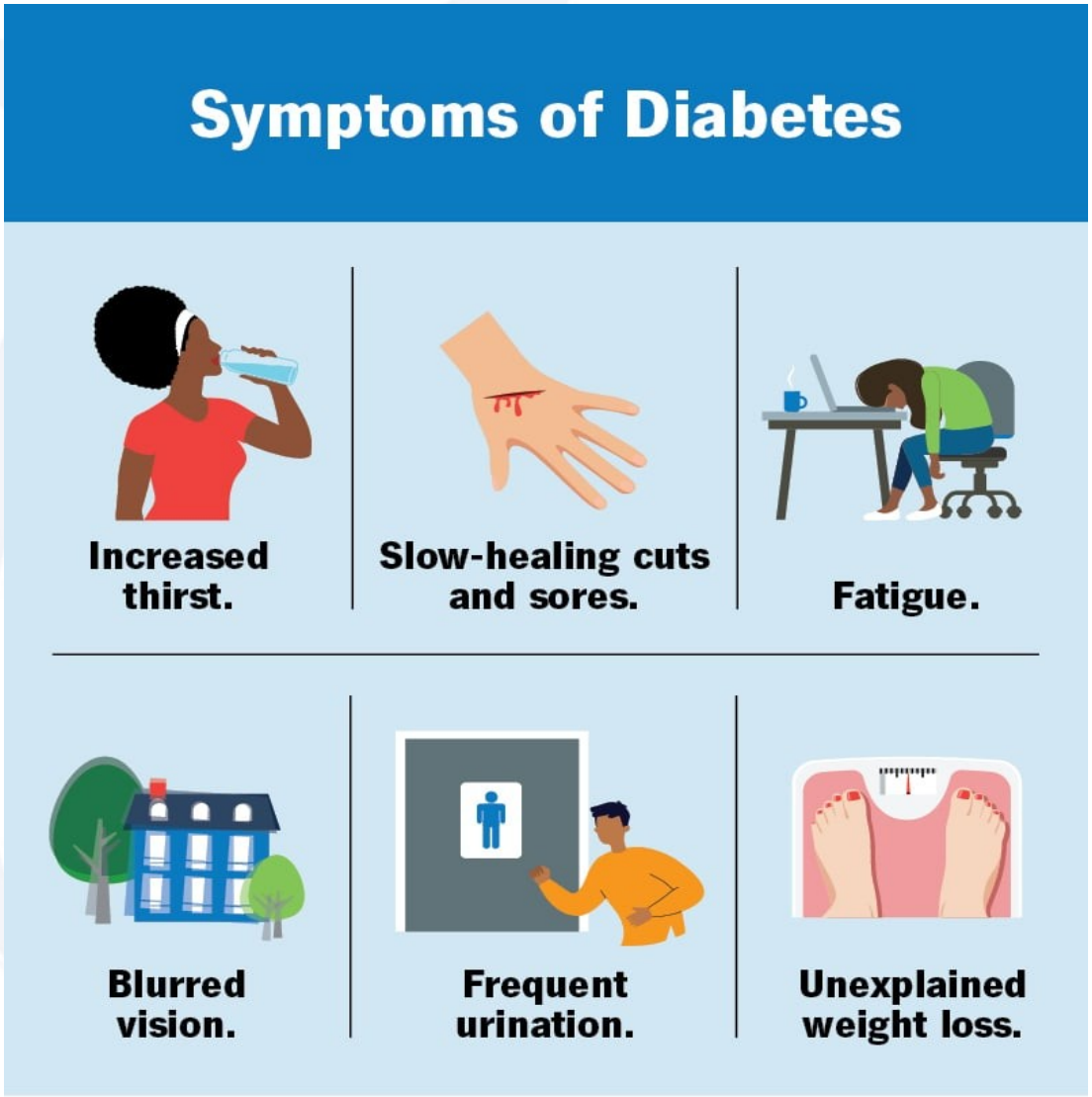
- **Accidental diagnosis**
- **Hyperglycemia symptoms**
- **acute or chronic complications**

Average or median clinical diagnosis delay of **2.2 to 2.4 years** (approx. 2–3 years) after the physiological onset of the disease.

Main clinical features of diabetes



Diabetes mellitus semiology



- Weight loss
- Polyuria+ polidypsia
- Blurred vision
- Nausea/ abdominal pain

- Paresthesia



DIAGNOSIS OF DIABETES

History :

➢ **Hyperglycemia review of symptoms:**

- **Polyuria :** due to glycosuria (osmotic diuresis)
- **Polydipsia :** due to polyuria

Weight loss : defect of insuline lipolysis

Hyperglycemia

 cell fasting state

 Compensatory polyphagia

polyuria+ polydipsia + polyphagia + weight loss

= Hyperglycemia symptoms

DIAGNOSiS OF DIABETES

Examination :

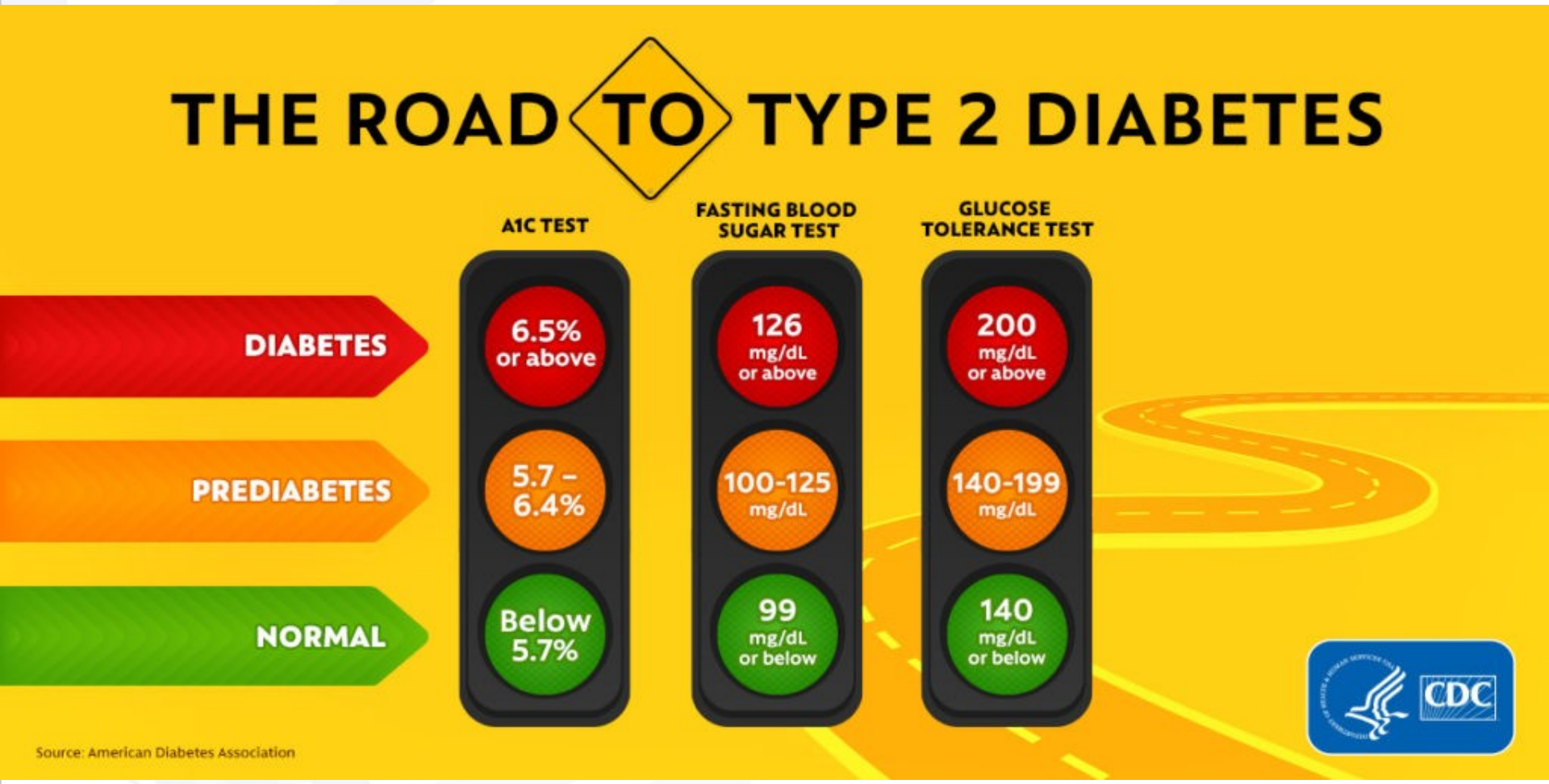
- **Capillary Blood Glucose:**

Measurement of glucose level in a capillary vessel

- **Glycosuria: As soon as the blood sugar level exceeds 1.8 g/L, the threshold of** tubular reabsorption of glucose, there is a passage of sugar in the urine.

- **Dehydration:** Signs of dehydration are sometimes found.
- **Weight/height with BMI calculation**
- **Infectious signs**
- **Examination for complications**

DIAGNOSIS OF DIABETES



DIAGNOSIS OF DIABETES

Criteria for diagnosis of diabetes

1.FPG ≥ 126 mg/dL (7.0 mmol/L) on 2 samples OR

2.2-h plasma glucose ≥200 mg/dL (11.1 mmol/L) during OGTT. OR

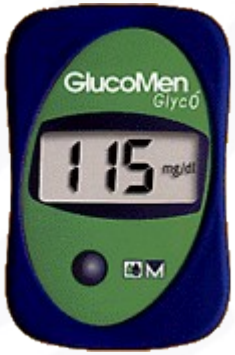
3.A1c ≥ 6.5% (48 mmol/mol) OR

4.In a patient with classic symptoms of hyperglycemia or hyperglycemic crisis, a random plasma glucose ≥200 mg/dL (11.1 mmol/L)

DIAGNOSiS OF DIABETES











DIAGNOSiS OF DIABETES

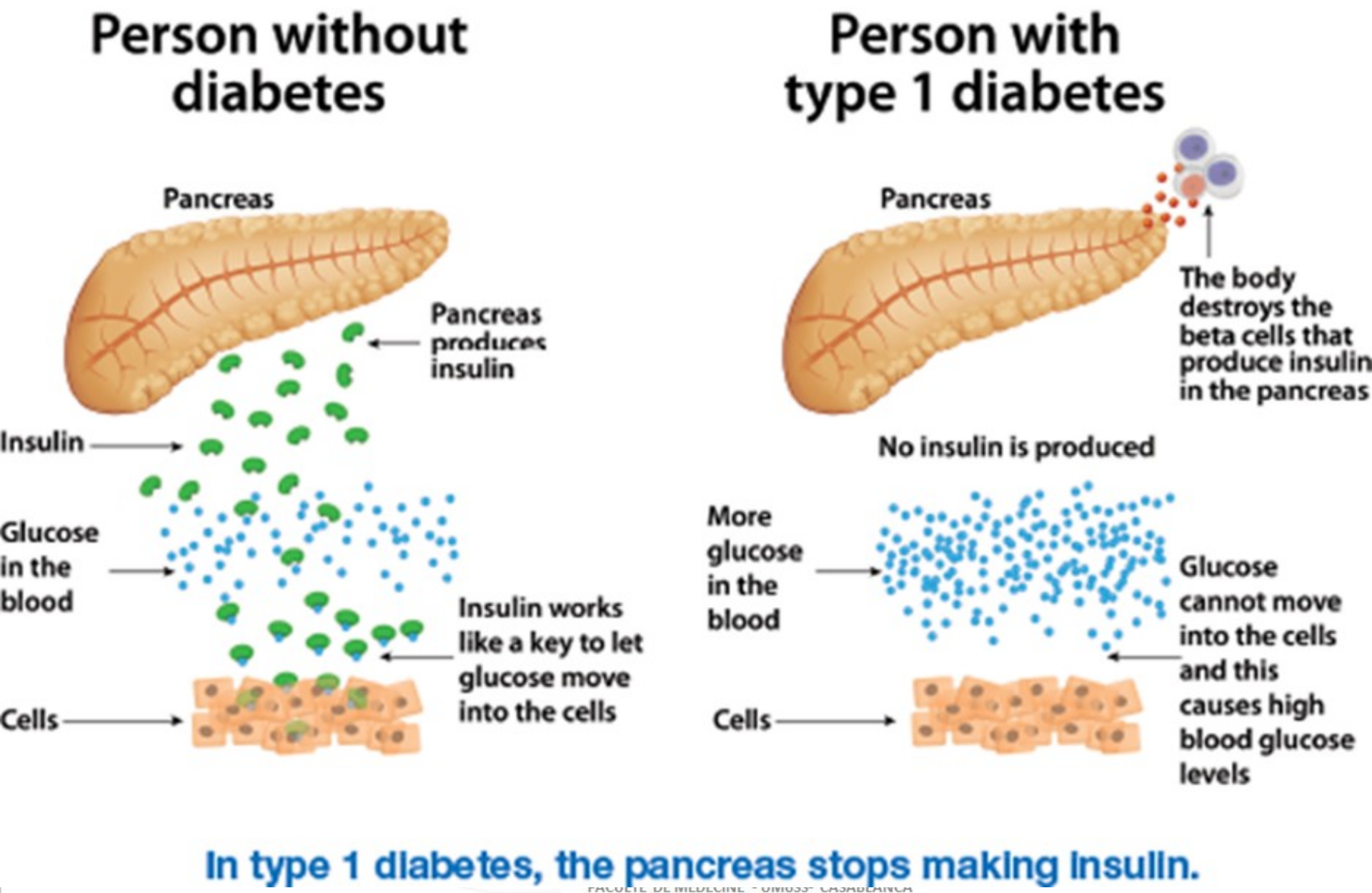
Urine test strip



Type 1 diabetes

- Total insulin deficiency
- Alteration of pancreatic secretion secondary to autoimmune disease
- Destruction of betapancreatic cells by autoantibodies
- Occurs after 90% of the betapancreatic cells have been destroyed

Type 1 diabetes



In type 1 diabetes, the pancreas stops making insulin.

Type 1 diabetes

Acute onset

Young subject (<40 yo)

Hyperglycemia symptoms

Body habitus : normal to thin

Diabetic Ketosis or ketoacidosis at onset

No Familial history of diabetes

History of auto-immune disease

No cardiovascular factors associated

Need for vital and definitive Insulintherapy

Chronic Complications assesment after

TYPE 2 DIABETES

➢ **Genetic factors:** history of diabetes

➢ **Environmental factors:**

- Nutritional imbalance

- Insufficient physical activity

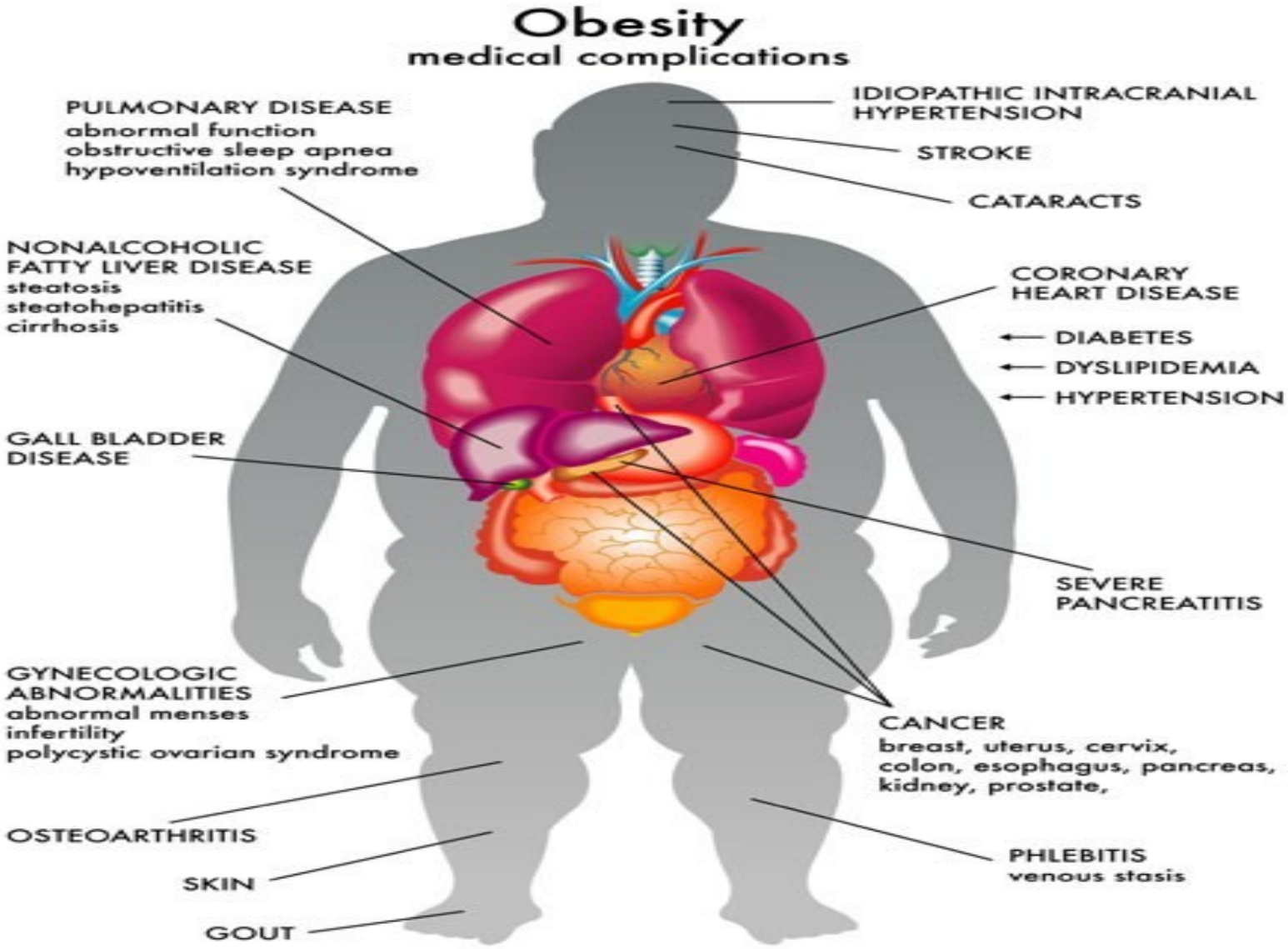
- Obesity, especially android obesity

➢ **Metabolic factors:**

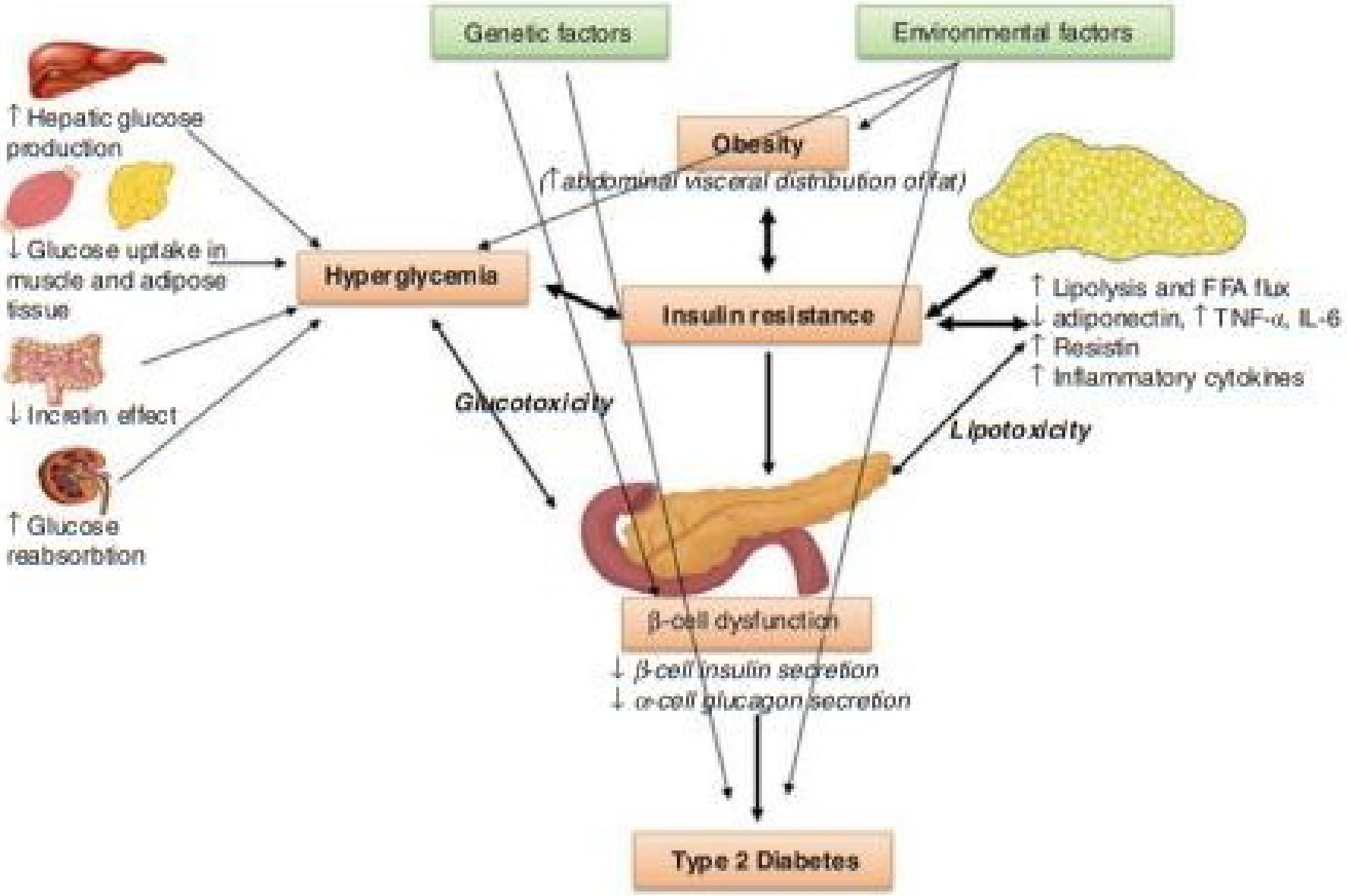
Insulin resistance

Progressive Insulin Deficiency

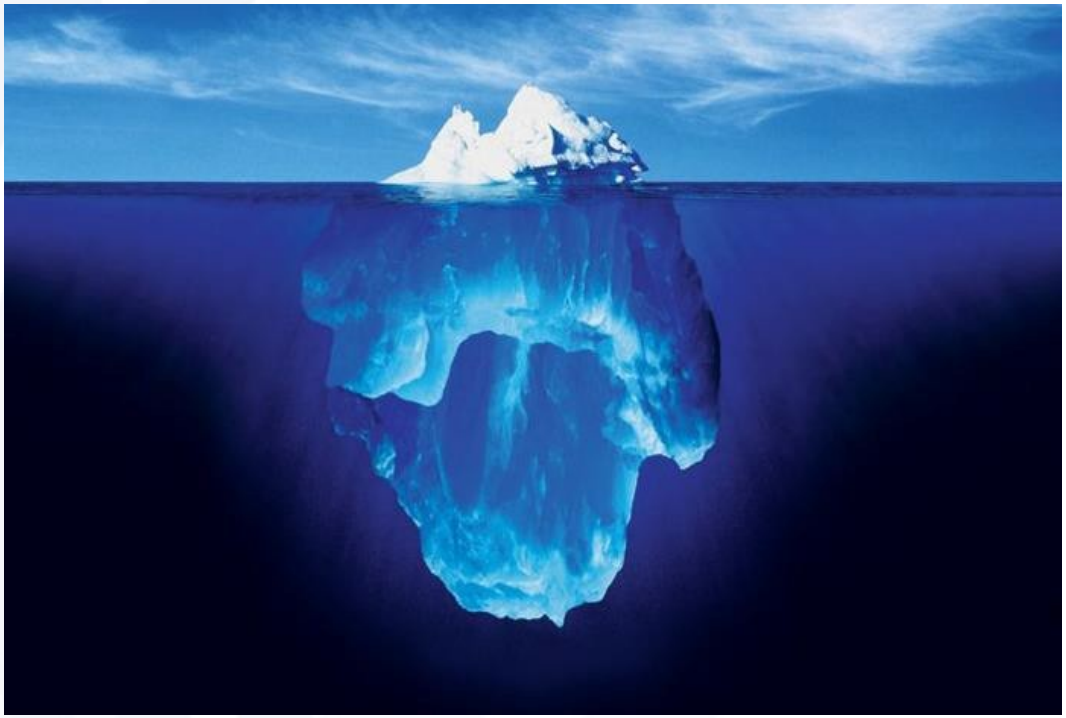
Obesity link- metabolic syndrome- type 2 diabetes



TYPE 2 DIABETES



TYPE 2 DIABETES



HYPERGLYCEMIA

Obesity
Hypertension
Cardiovascular disease
Kidney disease
Dyslipidemia

TYPE 2 DIABETES

- Often discovered incidentally or during chronic complications
- Older subject(>40 years)
- May be asymptomatic or moderately symptomatic
- Ketoacidosis if additional factor(infection, intercurrent disease)
- Hyperglycemia symptoms if very umbalanced
- Familial history of diabetes
- Overweight or obesity
- Cardiovascular risk factors
- Chronic Complications assesment at the discovery of diabetes

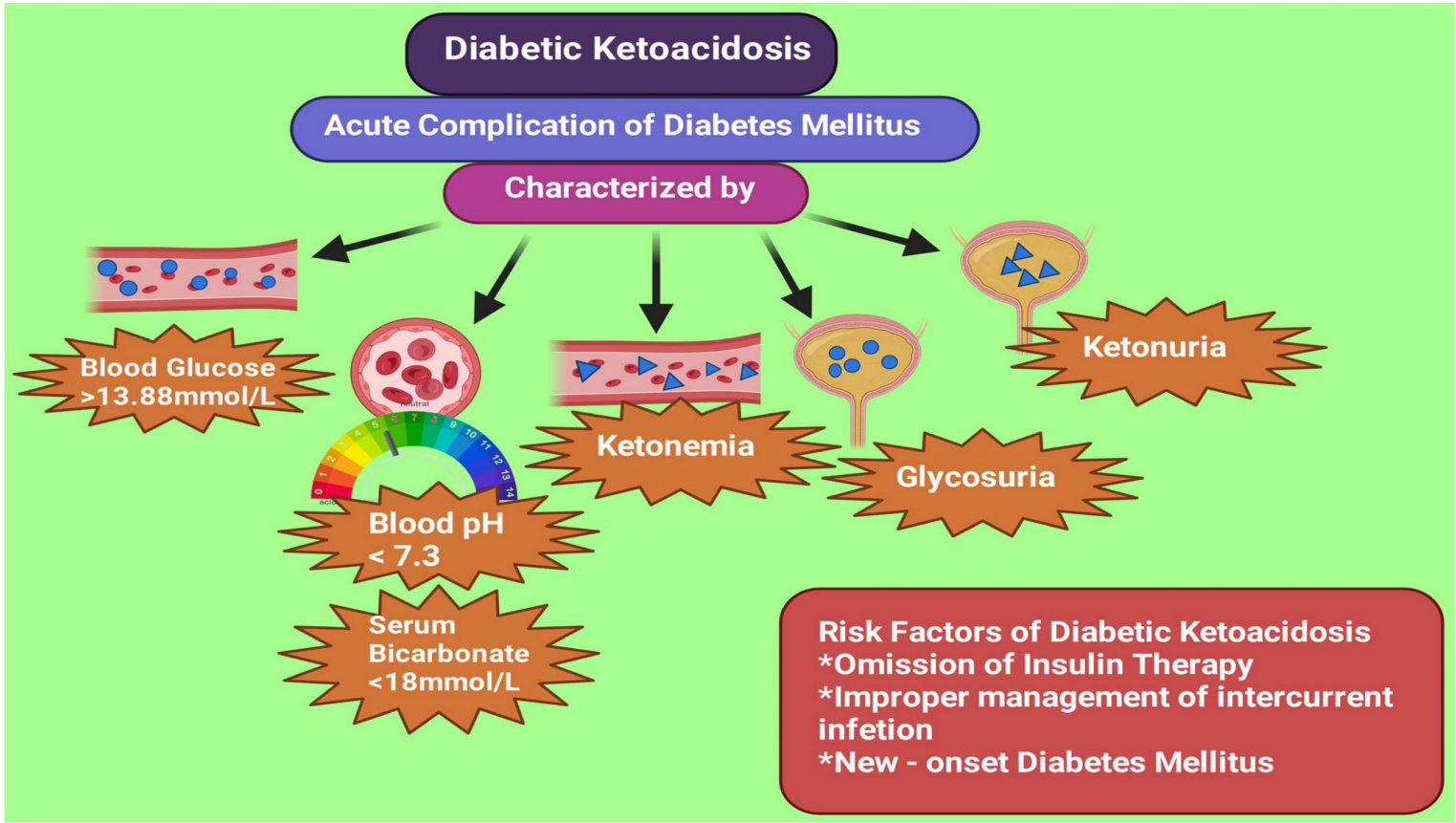
Classification of Diabetes Mellitus

Type of diabetes	Type 1 diabetes	Type 2 diabetes
Frequency	10-15%	85-90%
Familial history of DT	+	+++
Age of onset	Young	after 40 years
Start mode	acute	progressive
Overweight/obesity	absent	present
Hyperglycemia Symptoms	+++	-
Insulin secretion	Absent	persistent
Ketosis	frequent	absent
Associated autoimmune diseases	+	-
Self antibodies	+	-
CVx risk factors	-	+

COMPLICATIONS OF DIABETES

- 2 types of complications :

✓ Acute



COMPLICATIONS OF DIABETES

- **2 types of complications :**
  - ✓ **Chronic complications :**



COMPLICATIONS OF DIABETES

- 2 types of complications:

✓ Acute complications

-Diabetic ketoacidosis

-Hypoglycemia

-Hyperosmolar hyperglycemia state

-lactic acidosis

✓ Chronic Complications:

degenerative complications secondary to chronic hyperglycemia

COMPLICATIONS OF DIABETES

- **2 types of complications:**

✓ **Acute complications**

-Diabetic ketoacidosis

-Hypoglycemia

-Hyperosmolar hyperglycemia state

-lactic acidosis

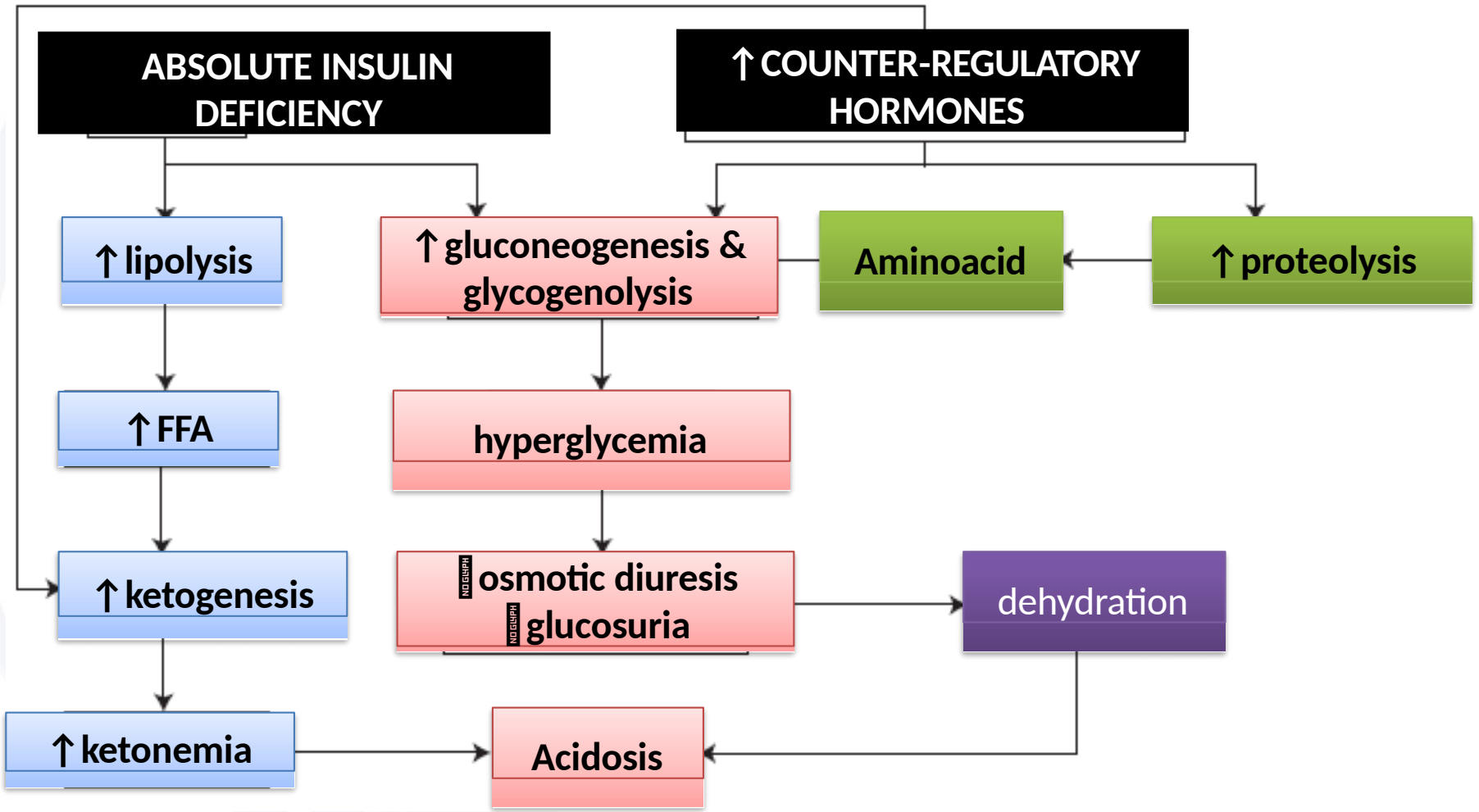
✓ **Chronic Complications:**

degenerative complications secondary to chronic hyperglycemia

ACUTE COMPLICATIONS OF DIABETES

Diabetic ketoacidosis

DIABETIC KETOACIDOSIS: PATHOPHYSIOLOGY



DIABETIC KETOACIDOSIS: CLINICAL PRESENTATION

- Gradual onset of signs and symptoms (a few days)
- Except in children, pregnant women and the elderly, or in the event of a malfunctioning insulin pump.
- classification: mild ketoacidosis, moderate to severe ketoacidosis.

- ***Mild ketoacidosis :***

➢ Review of symptoms:

- **Signs of hyperglycemia:** polyuria, polydispia and constant thirst, Blurry vision, Frequent urination at night,weight loss.
- **Signs of ketosis:** abdominal pain, nausea, vomiting, anorexia, ketonic breath (fruity breath).

DIABETIC KETOACIDOSIS: CLINICAL PRESENTATION

- ***Mild ketoacidosis :***

➢ Physical examination :

- Capillary blood glucose usually > 2.50 g/l
- Ketonuria with urine test strip
- High ketonemia (combined blood glucose meter) and ketonemia

DIABETIC KETOACIDOSIS: CLINICAL PRESENTATION

- *Moderate to severe Ketoacidosis*
- Physical examination :

- Worsening of pre-existing symptoms
- **Signs of acidosis:**

-**Dehydration:** predominantly in the extracellular sector (dry skin, arterial hypotension and tachycardia).

It may be associated with intracellular dehydration (thirst, Dry mucous membranes, and sunken eyes)

- **Kussmaul respiration** (tachypnea with deep inspirations)
- **Abdominal pain:** pseudosurgical
- **Altered mental status**, ranging from drowsiness to coma

DIABETIC KETOACIDOSIS: CLINICAL PRESENTATION

- *Moderate to severe Ketoacidosis :*

- Laboratory tests :

•Biochemical diagnostic criteria for diabetic ketoacidosis in adults

- **Ketonemia ≥3 mmol/L or significant ketonuria (>2+).**
- **Blood glucose >250 mg/dL (>14) mmol/L or known diabetes mellitus.**
- **Bicarbonate (HCO3 –) <7.3.**

Hypoglycemia

Hypoglycemia

- Abnormal decrease of blood glucose levels

- Defined by a blood glucose level ≤ 0.7g/l (ADA threshold)

- can have a dramatic effect on quality of life
- Increased morbidity and mortality

- Dreaded complication, directly or indirectly responsible for 2 to 4% of deaths due to diabetes

- **Emergency treatment +++**

- **Etiologies:** Insulin or sulfonyurea drugs overdose
  delay in taking a meal, Insufficient food intake
  Intense physical activity

Hypoglycemia

➢ circumstances of discovery

- Clinical symptoms of hypoglycemia
- Incidental, by self-monitoring of blood glucose: asymptomatic
- Complications: seizures , coma

➢ Clinical features:

Whipple Triad

- **Symptoms of hypoglycemia**
- **Low blood glucose (values<70mg/dl)**
- **Resolution of symptoms after glucose intake**

Signs and symptoms of hypoglycemia

Adrenergic (Autonomic)	Neuroglycopenic
tremor palpitations anxiety sweating hunger	confusion drowsiness speech difficulty weakness incoordination headache visual disturbance behavior changes convulsions loss of consciousness

HYPOGLYCEMIA

➢ **Clinical features:**

- **Symptoms of neuroglucopenia:**

✓ **Moderate:**

Dizziness
Drowsiness
Irritability
Confusion
Personality changes
Difficulty speaking
Weakness
Headache
Blurred vision
Impaired judgment

✓ **Severe:**

HYPOGLYCEMIA

➢ **Clinical features:**

Classification of hypoglycemia:

Level	Glycemic criteria	Criteria
Hypoglycemia alert value (level 1)	≤70 mg/dL (3.9 mmol/L)	Sufficiently low for treatment with fast-acting CH and dose adjustment of glucose-lowering therapy
Clinically significant hypoglycemia (level 2)	<54 mg/dL (3.0 mmol/L)	Sufficiently low to indicate serious, clinically important hypoglycemia Autonomic and neuroglycopenic symptoms present, often requiring external assistance for recovery
Severe hypoglycemia (level 3)	No specific glucose threshold	Hypoglycemia associated with severe cognitive impairment, seizures, coma requiring external assistance for recovery and injectable treatment (glucose, glucagon)

CHRONIC COMPLICATIONS OF DIABETES

- **Microvascular complications** → disease of the smallest blood vessels, the capillary and the precapillary arterioles, is manifested mainly by thickening of the capillary basement membrane ⇒

– **diabetic retinopathy**
– **diabetic nephropathy**
– **diabetic neuropathy**
– **cardiomegaly with heart failure**

CHRONIC COMPLICATIONS OF DIABETES

- **Macrovascular complications** → an accelerated form of atherosclerosis ⇒

–chronic heart disease (↑ incidence of myocardial infarction)
– cerebrovascular disease (stroke)
– peripheral artery disease (peripheral gangrene)

DIABETIC RETINOPATHY

- Hyperglycemia occlusion of the retinal vessels
- Retinal ischemia extends a reactive proliferation of neovessels

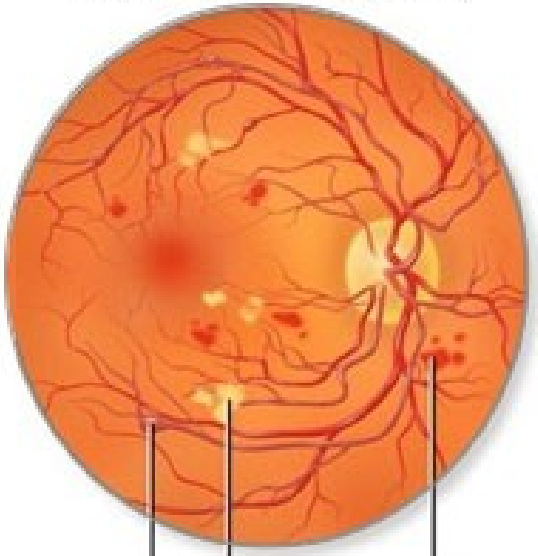
- **Non-proliferative retinopathy**
- **Severe or pre-proliferative retinopathy**
- **Proliferative retinopathy**

Poor clinical signs or asymptomatic screening of patients at risk

Fundus examination in diabetics +/- retinal angiography

TYPES OF DIABETIC RETINOPATHY

Non-proliferative diabetic retinopathy



Proliferative diabetic retinopathy



DIABETIC RETINOPATHY

Circumstances of discovery

- Vision is usually normal until vitreous hemorrhage or retinal detachment occurs.

DIABETIC RETINOPATHY

History: diabetes history, glycemic control (A1C) and blood pressure control, history of other chronic complications

Clinical examination

Ophthalmological examination:

- measurement of visual acuity with and without optical correction
- measurement of intraocular pressure (glaucoma test)
- Examination of the iris and the lens (for rubeosis iridis and cataracts)
- examination of the vitreous and the fundus after pupillary dilation

DIABETIC NEUROPATHY

-heterogeneous group of disorders with diverse clinical manifestations.

- Peripheral and autonomic neuropathy may be sub-clinical or symptomatic.

- Same frequency in T1 as in T2.

-High morbidity:

One million amputations each year

- 15–40 times more likely to require a lower-limb amputation compared to the general population

-Early detection and prevention

DIABETIC NEUROPATHY: PATHOPHYSIOLOGY

Hyperglycemia Two types of pathologic changes with diabetic peripheral neuropathies.

 ischemia of vessels that supply the nerves **vessel ischemia.**

 Segmental demyelinization process that affects the Schwann cell **slowing of nerve conduction.**



Histological and functional changes.

DIABETIC NEUROPATHY

- **Polyneuropathies (bilateral sensory)**

✓ Paresthesias, including numbness and tingling

✓ Impaired pain, temperature, light touch, two-point discrimination, and vibratory sensation

✓ Decreased ankle and knee-jerk reflexes

- **Mononeuropathies**

✓ Involvement of a mixed nerve trunk that includes loss of sensation, pain, and motor weakness.

- **Amyotrophy**

✓ Associated with muscle weakness, wasting, and severe pain of muscles in the pelvic girdle and thigh.

DIABETIC NEUROPATHY

- **Autonomic:**

- Postural hypotension
- Resting tachycardia
- Loss of sweating
- Gastrointestinal neuropathy : Gastroparesis ,Diabetic diarrhea
- Urinary bladder atony
- Erecticle dysfunction

Clinical assessment

1 Review of signs:

- Careful history
- Assessement of symptoms: Pain, Numbness,Tingling Weakness, Ataxia ,Upper-limb symptoms
- DN4 questionnaire, Toronto score …
- Autonomic neuropathy signs++: skin dryness, hyperkeratosis, loss of sweating

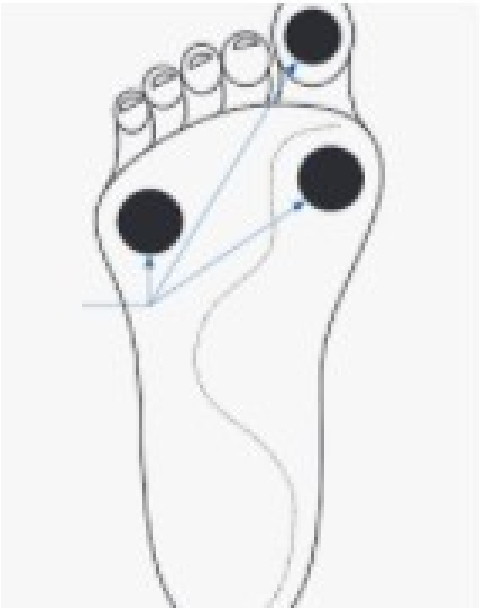
2-physical examination:

- Temperature and pinprick sensation
- vibration sensation using a 128-Hz tuning fork

DIABETIC NEUROPATHY

10g Monofilament Test





DIABETIC NEUROPATHY

Examination using 128-Hz tuning fork



DIABETIC NEUROPATHY

DN4 Questionnaire

INTERVIEW OF THE PATIENT

Question 1: Does the pain have one or more of the following characteristics?

	Yes	No
1 – Burning		
2 – Painful cold		
3 – Electric shocks		

Question 2: Is the pain associated with one of more of the following symptoms in the same area?

	Yes	No
4 – Tingling		
5 – Pins and needles		
6 – Numbness		
7 – Itching		

EXAMINATION OF THE PATIENT

Question 3: Is the pain located in an area where the physical examination may reveal one or more of the following characteristics?

	Yes	No
8 – Hypoesthesia to touch		
9 – Hypoesthesia to prick		

Question 4: In the painful area, can the pain be caused or increased by:

	Yes	No
10 – Brushing		

The total score is calculated as the sum of the 10 items and the cut-off value for the diagnosis of neuropathic pain is a total score of 4/10.

Total

DIABETIC NEUROPATHY

Toronto score

Toronto clinical neuropathy scoring system

Symptom scores	Reflex scores	Sensory test scores
Foot • Pain • Numbness • Tingling • Weakness Ataxia Upper-limb symptoms	Knee reflexes Ankle reflexes	Pinprick Temperature Light touch Vibration Position

Sensory testing was performed on the first toe. *Symptom scores*: present = 1; absent = 0. *Reflex scores*: absent = 2; reduced = 1, normal = 0. *Sensory test score*: abnormal = 1. Normal = 0. Total scores range from normal = 0 to maximum of 19.

Interpretation:

≤5 no neuropathy, 6–8 mild neuropathy,

9–11 moderate neuropathy, ≥12 severe neuropathy.

Active prevention of diabetic foot injury



DIABETIC NEUROPATHY

Reflexe assessment



DIABETIC NEUROPATHY

Characteristics of neuropathic feet

hot feet
Dry skin with hyperkeratosis
Superficial sensation disorders: monofilament+.
Deep (proprioceptive) sensation disorders (reflexes, tuning fork)
Deformities(hammer , claw toes)







Charcot arthropathy

severe neuropathy
long-standing diabetes
initiating event = injury,
causing bone fracture
gross deformity
cubic foot
acute Charcot foot



Diabetic Foot Ulcer



Autonomic neuropathy

✓ **Cardiac autonimic neuropathy**

✓ **Orthostatic hypotension:**

-Blood pressure measurement in supine position and after 3 minutes of orthostatism: loss of 30 mmHg of PAS and/or 10 mmHg at 3 minutes

✓ **Cardiac autonomic neuropathy**

-Permanent tachycardia, cardiac denervation: silent myocardial ischemia

-Vasomotor instability of the lower limbs

Clinical features

CoD:

- Systematic annual check-up: micro or macroangiopathy.
- Edematous syndrome, nephrotic syndrome.
- Presence of hypertension
- Proteinuria with strips.

Clinical presentation:

1. **History** : Type of diabetes, history of diabetes
   Urinary tract infection
   Glycemic balance
2. **Examination:**

✓ Asymptomatic disease
✓ Edema
✓ HTN
✓ Urine strip test

Clinical features

Assessment :

- Conventional 24-h urine collections
- Ratio of albumin (g/L) to creatinine (mg/L) ACR in an early morning spot urine
- Blood serum creatinine , glomerular filtration rate (GFR)

DIABETIC NEPHROPATHY:

SYMPTOMS, DIAGNOSIS, CAUSES & TREATMENT

SYMPTOMS

- Deteriorating blood pressure regulation
- Protein in the urine
- Swelling of ankles, hands feet or eyes
- Increased want to urinate
- Shortness of breath
- Loss of appetite
- Nausea and vomiting
- Persistent itching
- Exhaustion

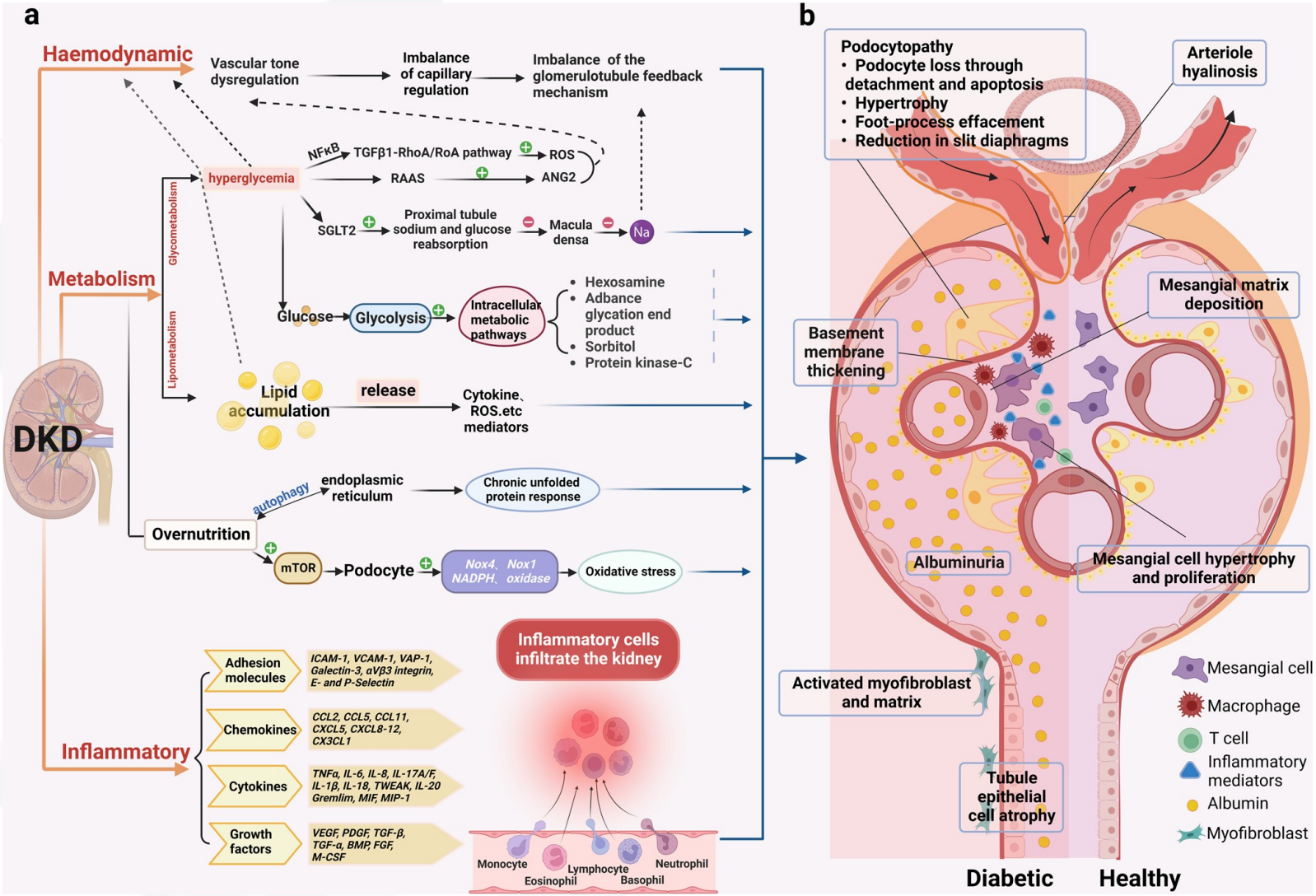


CAUSES

- Diabetes
- Prolonged Hyperglycemia
- Abnormal Blood Lipid Levels
- High Blood Pressure
- Poor Glycemic Control
- Ethnicity
- Smoking
- Obesity

advancells
THE POWER TO CURE

NEPHROPATHY- PHYSIOPATHOLOGY



UNDERSTANDING YOUR KIDNEY LAB TESTS



ACR (urine test)

- Albumin-to-creatinine ratio (ACR) is a measure of the protein albumin in your urine



eGFR (blood test)

- Estimated glomerular filtration rate (eGFR) is a measure of kidney function

@beyonddiabetesnutrition

DIABETIC NEPHROPATHY

Table 8.18 Prognosis of CKD by GFR and albuminuria categories: KDIGO 2012

GFR categories (mL/min/1.73 m²) description and range		
G1	Normal or high	≥90
G2	Mildly decreased	60–89
G3a	Mildly to moderately decreased	45–59
G3b	Moderately to severely decreased	30–44
G4	Severely decreased	15–29
G5	Kidney failure	<15
Persistent albuminuria categories, description, and range		
A1	A2	A3
Normal to mildly increased	Moderately increased	Severely increased
ACR <30 mg/g ACR <3 mg/mmol	ACR 30–300 mg/g ACR 3–30 mg/mmol	ACR > 300 mg/g ACR > 30 mg/mmol
AER <30 mg/24 h	AER 30–300 mg/24 h	AER > 300 mg/24 h

ACR albumin-to-creatinine ratio, *AER* albumin excretion rate, *CKD* chronic kidney disease. Nephrotic syndrome (albumin excretion usually >2200 mg/24 h [ACR > 2200 mg/g; >220 mg/mmol]).

DIABETIC MACROANGIOPATHY

B- Peripheral Artery Disease (PAD):



Dorsal Pedal Pulse



Posterior Tibial Pulse

DIABETIC MACROANGIOPATHY

B- Peripheral Artery Disease (PAD) :

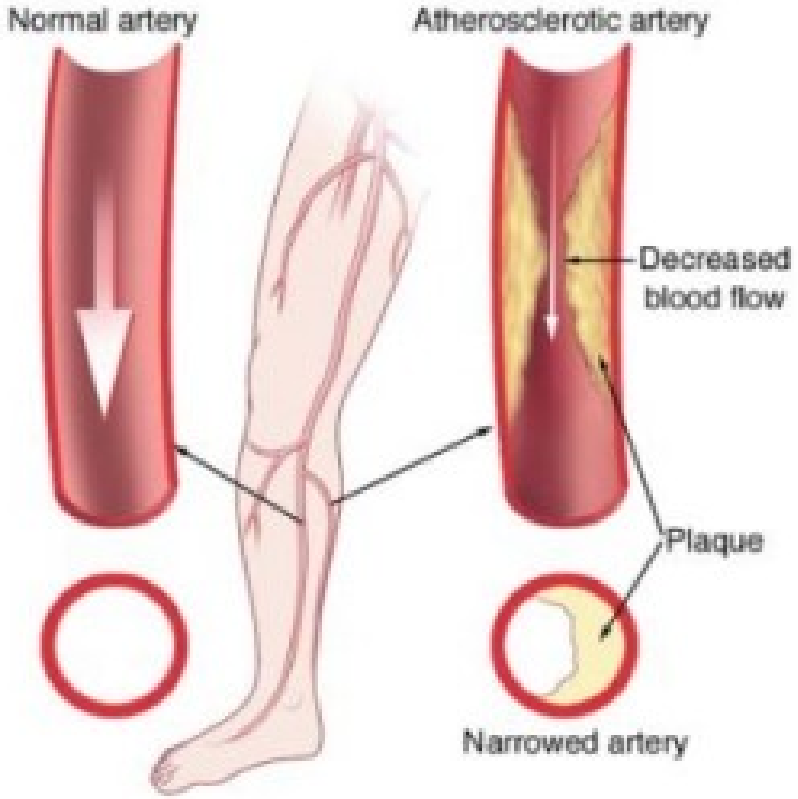
C-Clinical examination:

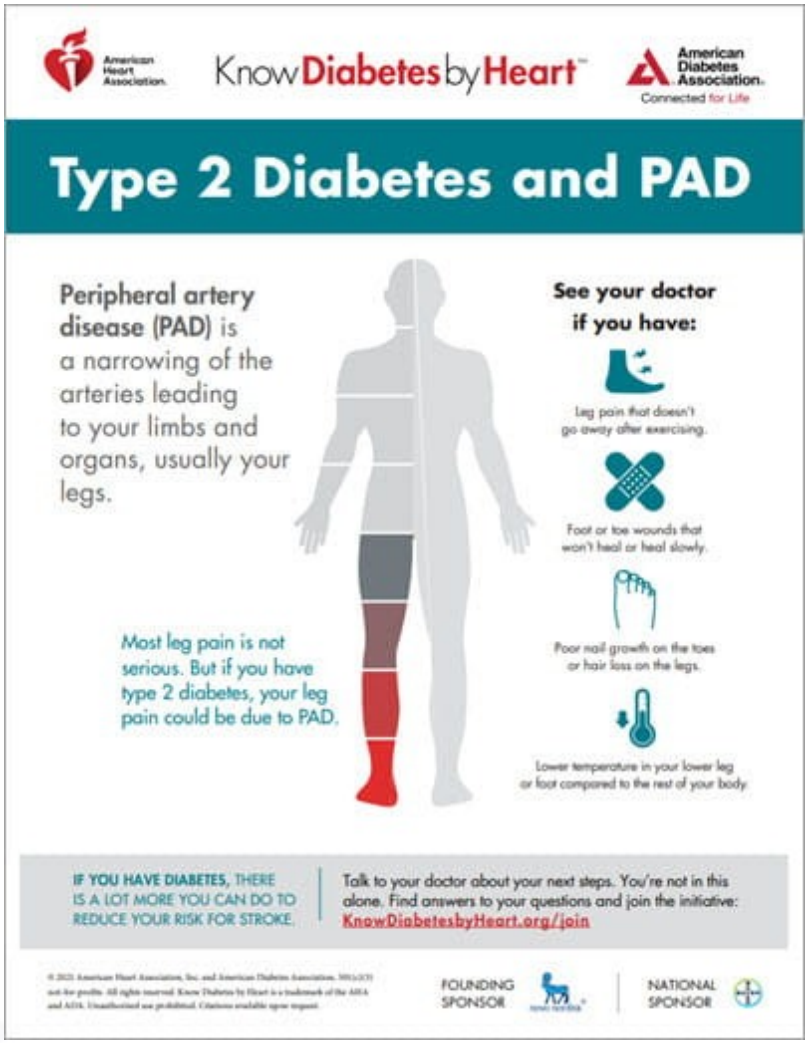
- Loss of pulses
- Pain on palpation
- Pallor
- Muscle atrophy and loss of hair
- Cool and cyanotic skin
- Presence of bruit
- brittle, slow-growing toenails

. D-Paraclinical examination:

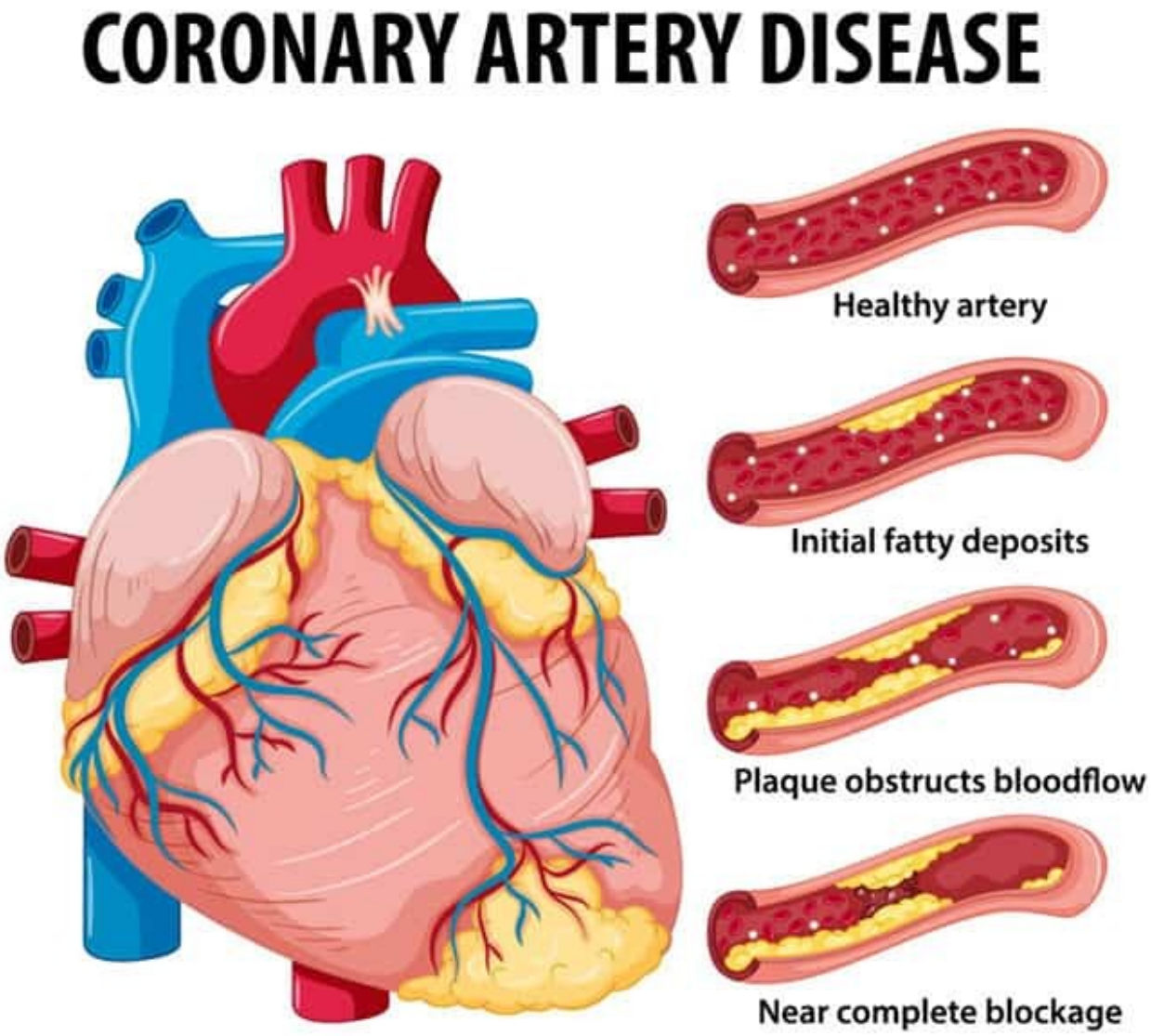
- ankle-brachial index (ABI)
- Arterial Doppler of the lower limbs
- CT angiography and MRA if if the patient is a candidate for angioplasty or by pass surgery.

Peripheric artery disease





Diabetic coronary disease



DIABETIC MACROANGIOPATHY

Coronary Artery Disease (CAD)

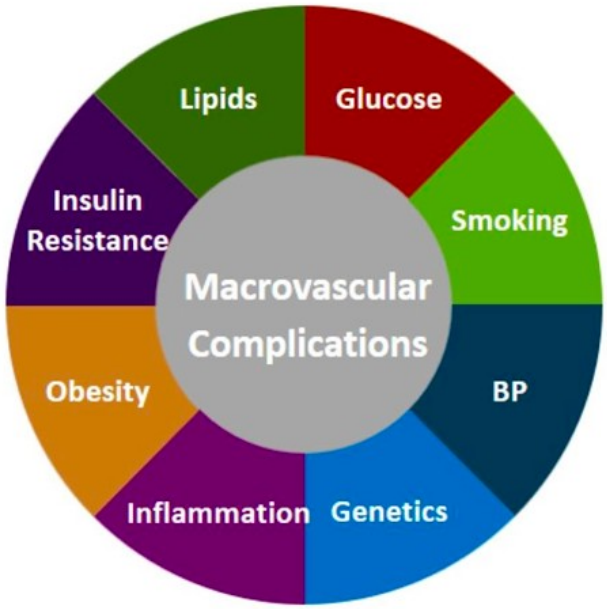
A- CoD:

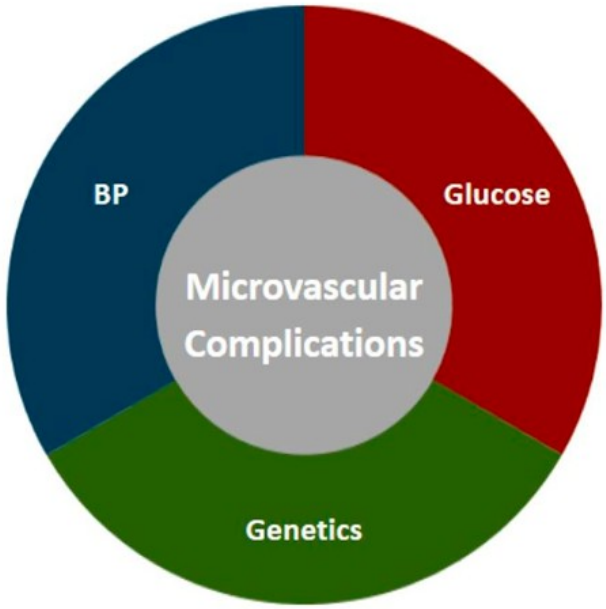
- **Atypical chest pain:** Epigastric ache, exertional dyspnea, palpitations, lipothymia
- Typical chest pain (uncommon)
- silent myocardial ischemia (SMI),

- **Acute complications of diabetes or glycemic imbalance**
- **Complications:** Heart failure

B-History:

Age, gender, family history, cardiovascular risk factors, describe chest pain, look for other macroangiopathies

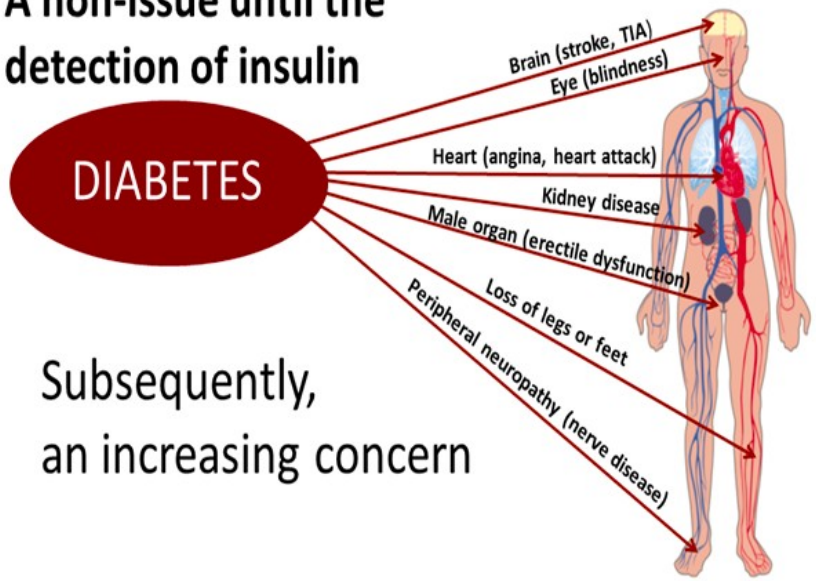




Circulation. 2004;1029.
Diabetes Care. 2012;35:556-564
Clin Diabetes. 2008;26:77-82

Vascular Injury and Diabetes

A non-issue until the detection of insulin



Subsequently,
an increasing concern

Glycemic Control and Macrovascular Disease

ACCORD

The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812 JUNE 12, 2008 VOL. 358 NO. 24

Effects of Intensive Glucose Lowering in Type 2 Diabetes

The Action to Control Cardiovascular Risk in Diabetes Study Group*

ADVANCE

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Intensive Blood Glucose Control and Vascular Outcomes in Patients with Type 2 Diabetes

The ADVANCE Collaborative Group*

VADT

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Glucose Control and Vascular Complications in Veterans with Type 2 Diabetes

William Duckworth, M.D., Carlos Abraira, M.D., Thomas Moritz, M.S., Domenic Reda, Ph.D., Nicholas Emanuele, M.D., Peter D. Reaven, M.D.,

ACCORD Study Group. *N Engl J Med*. 2008;358:2545-2559.
The ADVANCE Collaborative Group. *N Engl J Med*. 2008;358:2560-2572.
Duckworth W, et al. *N Engl J Med*. 2009;360:129-139.

DIABETIC MACROANGIOPATHY

diabetic coronary artery disease

C-Clinical examination:

-Rule out other causes of chest pain

-Cardiac and vascular auscultation

D-Paraclinical examination:

- ECG

-Troponin assay: myocardial necrosis

-Coronary angiography

DIABETIC MACROANGIOPATHY

Gangrene



DIABETIC MACROANGIOPATHY

Obliterative arteriopathy of the lower limbs:



DIABETIC MACROANGIOPATHY

Stroke

Strokes are more frequent in diabetics

ischemic stroke+++

a stroke can :

- ✓ reveal diabetes
- ✓ induce an DKA revealing diabetes
- ✓ Assessment: ECG, echocardiography/cardiac stress test, coronary angiogram, carotid Doppler ultrasonography, CT scan.

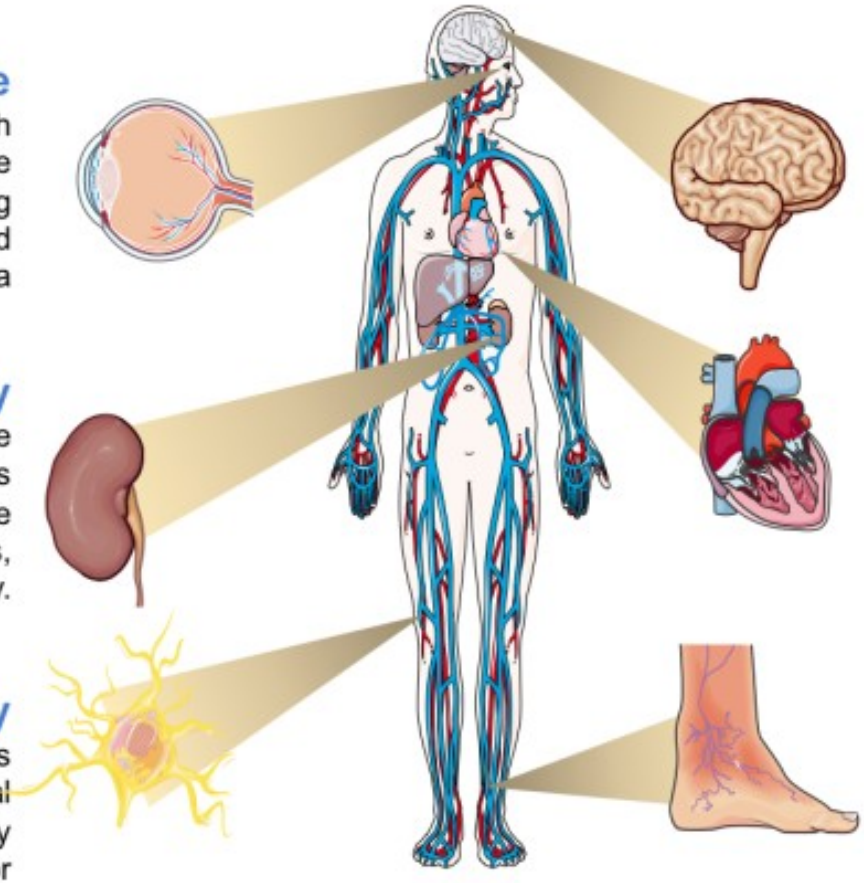
Major Complications of Diabetes

Microvascular

Eye
High blood glucose and high blood pressure can damage eye blood vessels, causing retinopathy, cataracts and glaucoma

Kidney
High blood pressure damages small blood vessels and excess blood glucose overworks the kidneys, resulting in nephropathy.

Neuropathy
Hyperglycemia damages nerves in the peripheral nervous system. This may result in pain and/or numbness. Feet wounds may go undetected, get infected and lead to gangrene.



Macrovascular

Brain
Increased risk of stroke and cerebrovascular disease, including transient ischemic attack, cognitive impairment, etc.

Heart
High blood pressure and insulin resistance increase risk of coronary heart disease

Extremities
Peripheral vascular disease results from narrowing of blood vessels increasing the risk for reduced or lack of blood flow in legs. Feet wounds are likely to heal slowly contributing to gangrene and other complications.

CONCLUSION

- Chronic hyperglycemia secondary to a lack of secretion and/or action of insulin

- The diagnosis is based on specific criteria

- Type 1 diabetes: sudden onset, in a young subject, without overweight or obesity and without cardiovascular risk factors, ketosis.

- Type 2 diabetes: older subject, familial history of diabetes, overweight or obesity, progressive onset, sometimes during a complication, no inaugural ketoacidosis

- Severity of diabetes: acute and chronic complications